2025 International Conference on Data Science, Agents, and Artificial Intelligence (ICDSAAI 2025)
28-29 March 2025, Chennai, India

A Review of Generative AI in Recommendation Systems

Shivaani V P
Department of Computer Science and Engineering, Chennai Institute of Technology, Chennai, Tamilnadu-600069
shivaanivp.cse2023@citchennai.net

Muthupandi G
Department of Computer Science and Engineering, Chennai Institute of Technology, Chennai, Tamilnadu-600069
muthupandig@citchennai.net

S.Pavithra
Department of Computer Science and Engineering, Chennai Institute of Technology, Chennai, Tamilnadu-600069
pavithras@citchennai.net

2025 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI) | 979-8-3315-3755-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDSAAI65575.2025.11011809

Abstract — Traditional machine learning algorithms have paved the way for more sophisticated models mainly in the field of Gen AI. This review compares how Gen AI is used in recommendation systems to traditional AI methods and the application of both in various fields. Generative AI has transformed the systems for recommendation by correcting the flaws of conventional machine learning approaches. This review compares the use of Gen AI and conventional AI in RS throughout various fields, with a focus on GANs and VAEs. These generative models indicate their capabilities and challenges involving low data, cold start, and lack of diversity in recommendations. They also show potential limitations for further research. The report outlines recent works to represent how Gen AI enhances RS performance traces key trends, and addresses relevant emerging concerns with reliability and ethics. The results highlight hybrid approaches, promising further developments of the effective and flexible RSs. To conclude, the paper advocates for further research to be performed.

Keywords — Recommendation Systems, Collaborative Filtering, Content-based Filtering, Hybrid Approach, Generative Adversarial Networks (GANs).

I. INTRODUCTION

The main ability of machine learning is to extract knowledge from a data set and keep improving it over time. Several recent studies have shown that this technology is very good at image analysis, speech recognition, and recommendation systems it builds sophisticated user experiences tailored to individual preferences. Supervised, unsupervised, and reinforcement learning are the three prime approaches of machine learning [1]. Of the given models, each has a different purpose for modern artificial intelligence applications. As shown in the current review, supervised learning deals with training models on input-output pairs of labeled data so that the models become better for making predictions over tasks.

Conversely, unsupervised learning, with its basis on unlabeled data in uncovering hidden patterns and structures, therefore finds broad applications in customer segmentation and pattern recognition. Reinforcement learning is a unique paradigm wherein systems gain knowledge through feedback and interactions in the environment [2]. Recent advances in reinforcement learning have held special promise for adaptive systems needing to change to meet the needs of evolving users, as they enable ongoing optimization based on real-world feedback and interactions.

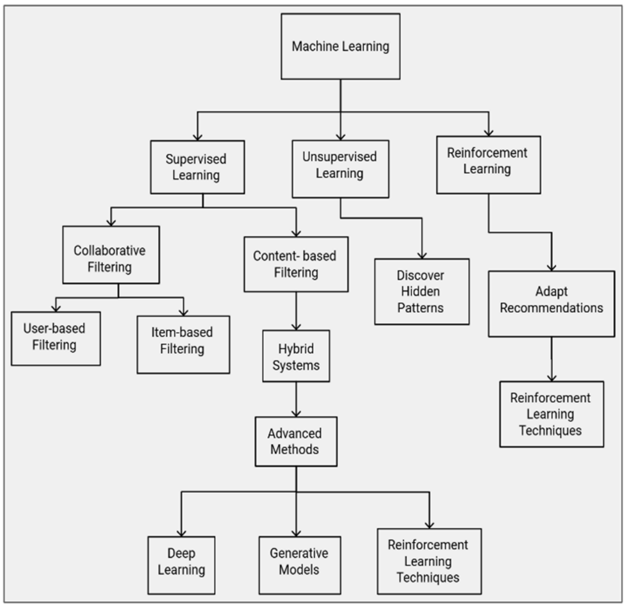

Figure 1: Machine learning techniques

Machine learning techniques are discussed below in figure 1. These systems provide personalized recommendations through algorithms that use machine learning like collaborative filtering, content-based filtering, as well as hybrid models.

Collaborative filtering is based on capturing the interaction of users and recommending items through the extraction of similarities from other users or items the user has interacted with. Despite this, challenges are still present, starting from cold start issues to data availability and scaling issues.

Content-based filtering involves recommending items similar to what a user is already interested in, using the characteristics of the items such as recommending jazz music to a person who happens to like jazz. It reduces dependence on other data, but it could result in filter bubbles in which recommendations become biased and less diversified.

II. LITERATURE SURVEY

Recommender systems can be broadly categorized into three types that have their methodology and strengths, which are: content-based filtering, collaborative filtering, and hybrid methods [3]. This is an important approach in giving personalized recommendations to customers within several industries, including entertainment and e-commerce. Techniques aimed at addressing special user needs and concerns with regard to data are served by these techniques. The Types of Recommender systems are discussed below in figure 2.

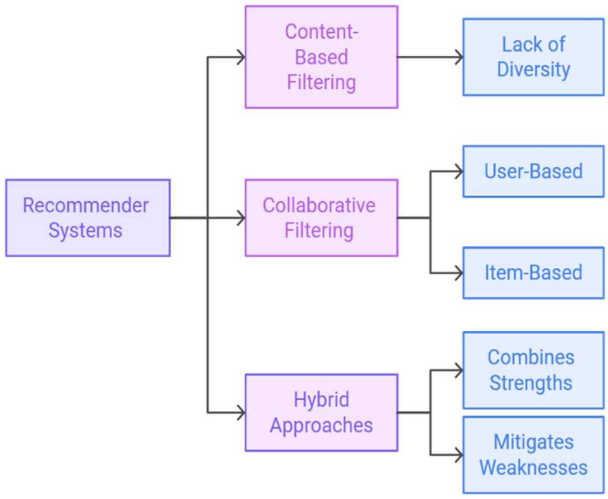

Figure 2 : Types of Recommender systems

1. Content-based Filtering:

This approach recommends items based on the attributes of those items and the interests of users. That is efficient for a person with peculiar tastes since this does not rely on other user's preferences. The major shortcomings of this approach can be pointed out as a lack of diversity in recommendations and dependence of recommendations on rich item features, which might not be available always [4].

2. Collaborative Filtering:

It comes highly recommended based on the similarities of other users. For personal recommendation and discovery, it works well with new content. However, it falls short in handling the "cold start" problem during recommendations to new users and sometimes is hurt by the sparsity of data in cases when only a few interactions exist between users and items [5].

3. Hybrid Approach:

Hybrid approaches are effective when the environment contains a huge amount of data and diversified preferences of users. It is highly resistant to sparsity of data and can include multiple sources such as ratings and reviews. Modern hybrid models use machine learning to adapt to dynamics and optimize the recommendation [6].

Table 1: Types of machine learning techniques

Techniques	Description	Advantages	Disadvantages	Example
Collaborative filtering	Analyses the interactions between the user and the item to find similar users	Suggests novel items	Suffers from cold start issues	Product recommendations in retail
Content-based filtering	Recommends items based on the item's features and the user history	Independent of user interaction	Limited diversity may cause a filter bubble	Recommending movies by genre
Hybrid approach	Combines content-based and collaborative methods for more accurate recommendations	Balances limitations of each method	Computationally complex	Social media feed personalization

2.1 Data Sparsity

Another significant problem in collaborative filtering for RS is data sparsity. This situation happens when there is less interaction between users and the items which results in sparse data and various missing values. It is therefore, hard for the system to detect any patterns or similarities. For instance, if the movie recommendation system has thousands of films but each of the users only rates a few films the matrix will be very sparse. This prevents meaningful relationships between movies from being identified by the algorithm [7]. Solutions such as matrix factorization and data augmentation are thus required to fill in the gaps and improve recommendations.

2.2 Scalability Issues

Scalability is another large challenge to the traditional approach to AI in RSs since the size of both users and items tends to scale rapidly. Recommender systems deal with extremely large datasets online, requiring very high computational capabilities. Traditional methods often incur problems with scaling since comparisons are to be done in pairs, making them very computationally expensive since as the data scale it becomes intractable [8]. This problem gets worse in dimensions where such systems have high dimensional data or where new users or items are added at a rate as the system continues updating its similarity calculations.

Techniques employed include distributed computing frameworks and the use of NoSQL databases, which help manage large scale data with the help of distributing the computational load across several nodes.

2.3 Cold Start Problem

A cold-start problem may thus arise when a recommender system has not obtained enough initial data about new users or items Interaction data on new items limits the ability of the system to relate it with existing items or users. This becomes quite challenging in systems that depend on interactions over time for inference in collaborative filtering. Often, hybrid approaches that combine the content-based technique and collaborative filtering are used to overcome this problem [9]. For example, content-based approaches can take advantage of an item's metadata to make initial recommendations. In some scenarios, the system may leverage data from similar users or exploit demographic data so that it provides initial recommendations until enough interaction data is gathered.

2.4 Limited Diversity

Another issue with traditional recommendation schemes is diversity in recommendations. In this respect, as systems attempt to personalize suggestions closely aligned with a user's preferences, it may lead to a lack of diversity, mostly termed the filter bubble. The system continues recommending items similar to the ones that the user has already interacted with and limits their exposure to the new content they might find interesting. This issue mainly arises in content-based recommendation approaches, where the system recommends items based on common attributes with the items that have been liked before yet it does not take into account more general preferences [10]. Diversity promotion requires hybrid approaches introducing novelty to the recommendations or techniques whose primary purpose is exactly to diversify the suggestions, balancing relevance against diversity of items.

2.5 Computational Complexity

Deep learning models often require a large amount of training data and a tremendous amount of computational power for processing high-dimensional data. Hence, recommendation systems remain challenging to implement in RS applications. For instance, neural collaborative filtering applies deep neural networks for the detection of complex patterns between user-item interactions. Its application requires many computational resources and hence cannot be practically processed by traditional AI algorithmic systems in real time, especially if no proper infrastructure is adopted. Data storage optimization and dimensionality reduction techniques are needed to tackle these challenges. Some key challenges associated with the approach include sparsity of data, scalability issues, the cold start problem, low item diversity, and high computational complexity [11]. These problems require hybrid approaches, big data platforms, and optimization techniques for high accuracy in recommendation and efficiency and usability in domains like social networks, movies, music, and books.

This review will explain in detail the major generating models applied in recommender systems, with special emphasis given to generative adversarial networks, variational autoencoders, and other generative AI techniques. Each of these models possesses specific properties that contribute to the effectiveness of the recommendations through synthetic data generation and handling familiar issues like data sparsity, user cold starts, and the lack of content diversity.

III. Generative AI Techniques in Recommender Systems

3.1 Generative Adversarial Networks (GANs)

GANs apply an adversarial training framework through a generator and a discriminator. The former generates synthetic data that is supposed to resemble the original interactions of the users with the items, while the latter evaluates the authenticity of those interactions. GANs employ adversarial training with the two opposing neural networks, the generator, and discriminator, for playing a zero-sum game. Iterative improvements based on adversarial feedback by GANs can produce highly realistic interactions that are valuable in recommendation contexts with the problem of sparsity in data. These networks are instrumental in the creation of synthetic user profiles and item interactions, hence enabling recommender systems to raise prediction accuracy when there is little available user data [12]. Furthermore, GANs have also proven to enhance recommendation diversity since they can imply new interaction types that might be interesting for the users. This in turn is turning the ability to go beyond known preference in neutralizing the so called "filter bubble" effect very often encountered in traditional recommender systems.

3.2 Variational Autoencoders (VAEs)

VAEs are yet another huge class of generation models applied to recommender system tasks in the collaborative filtering domain. Unlike regular auto-encoders, VAEs are also introduced to learn probabilistic representation over interactions in the user space. It can be described in three steps first, encoding data into a much lower latent space then, generating a new interaction by sampling from that space, and finally this encoded lower-dimension latent space becomes an opportunity to pick up complex patterns in user preferences, and it can even deal with missing data effectively when user interaction data is sparse and incomplete [13]. Recommendations can be made in VAEs based on inferred preferences both to new users and to items with few previous interactions. Furthermore, VAEs are much faster to train than GANs, thus suitable for real-time applications or fast updating applications.

3.3 Hybrid Models: Combining GANs and VAEs

Some of the advanced recommender systems leverage the combination of GANs and VAEs to combine the best of both worlds. These hybrid models can well overcome thold start problem and are most useful for cross-domain recommendations. As an example, a hybrid model can use a VAE that encodes user preferences in a latent space and then uses a GAN for generating synthetic data that enriches the training set with better recommendation diversity. GANs with VAEs significantly improve the generalization capabilities of the system in many recommender scenarios and result in robust solutions for personalization and novelty in suggestions [14]. All the recommender systems are shown in the following figure 3.

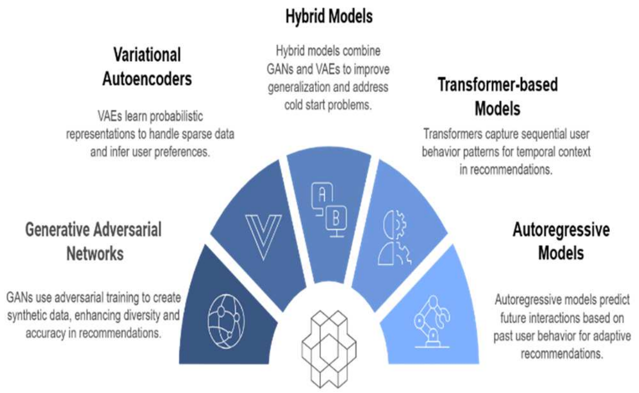

Figure 3: Recommender Systems

3.4 Transformer-based Models

Transformers are yet another model in the emergent generative AI field that, given their efficiency when handling sequences, have recently gained increased focus in state-of-the-art recommender systems. Most importantly, transformers were designed specifically to process sequential natural language but ultimately proved optimal in the capture of time patterns in user behavior. Modeling user interactions as sequences, transformers are very effective for predicting the outcome of future interactions that are based on past ones, which is crucial in applications like video or content recommendations where temporal context plays a huge role. It has proven to assign different weights to different past interactions, which makes it possible to capture long-term dependencies and hence highly relevant recommendations [15]. These models are specifically helpful for applications where it is known that the sequence of interactions for instance, sequences of views or purchases must be known to make accurate predictions.

3.5 Autoregressive Models

Autoregressive models, like those used in applications of text generation, have also been adapted for use in recommender systems. Based on the previous elements, these models predict the next element in a sequence. In this way, they are well suited for sequential recommendation tasks. For instance, one could use such an autoregressive model in an e-commerce website so that it would recommend items based upon those that are immediately preceded by one's previous purchases. Iterating through different user interactions and predicting what a user is likely to perform next, such models build a continuous recommendation loop that adapts as users continue interacting with different items over time [16]. Thus, the autoregressive models are especially useful in environments that relate to dynamic user intent and offer very responsive functionalities in response to changes in user behavior.

IV. CONCLUSION

In conclusion, the emergence of generative AI has shifted the entire general landscape of RS and has been a gateway for new approaches toward making the right amendments concerning user experiences across multiple platforms. The infusion of techniques based on generative AI into RS extends far beyond just following very basic methods to make recommendations much more personalized. These statements involved one of the biggest discoveries that the generative AI can leverage tremendous heterogenous datasets-user interactions, preferences, and even external content to tailor close suggestions improve precision, and enhance user engagement by delivering close-to-personalized content. These further empower generative AI to build artificial data and even remove cold start problems because it can be used to simulate user behavior and preferences. The results show the significance of incorporating user feedback into the generation process to continue optimizing algorithms with live interactions toward higher relevance in suggestions and building user trust and satisfaction. More advanced with the addition of generative AI and sentiment analysis, where generative AI is due to explain the interpretation of user sentiments from reviews into some nuanced understanding of preferences.

REFERENCES

[1]. Talaei Khoei, T., & Kaabouch, N. (2023). Machine Learning: Models, Challenges, and Research Directions. *Future Internet*, *15*(10), 332.

[2]. Wang, X., Zhao, Y., Qiu, C., Hu, Q., & Leung, V. C. (2024). Socialized Learning: A Survey of the Paradigm Shift for Edge Intelligence in Networked Systems. *IEEE Communications Surveys & Tutorials*.

[3]. Torkashvand, A., Jameii, S. M., & Reza, A. (2023). Deep learning-based collaborative filtering recommender systems: A comprehensive and systematic review. *Neural Computing and Applications*, *35*(35), 24783-24827.

[4]. Javed, U., Shaukat, K., Hameed, I. A., Iqbal, F., Alam, T. M., & Luo, S. (2021). A review of content-based and context-based recommendation systems. *International Journal of Emerging Technologies in Learning (iJET)*, *16*(3), 274-306.

[5]. Fkih, F. (2022). Similarity measures for Collaborative Filtering-based Recommender Systems: Review and experimental comparison. *Journal of King Saud University-Computer and Information Sciences*, *34*(9), 7645-7669.

[6]. Çano, E., & Morisio, M. (2017). Hybrid recommender systems: A systematic literature review. *Intelligent data analysis*, *21*(6), 1487-1524.

[7]. Kuo, R. J., & Li, S. S. (2023). Applying particle swarm optimization algorithm-based collaborative filtering recommender system considering rating and review. *Applied Soft Computing*, *135*, 110038.

[8]. Etemadi, M., Abkenar, S. B., Ahmadzadeh, A., Kashani, M. H., Asghari, P., Akbari, M., & Mahdipour, E. (2023). A systematic review of healthcare recommender

systems: Open issues, challenges, and techniques. *Expert Systems with Applications*, *213*, 118823.

[9]. Panda, D. K., & Ray, S. (2022). Approaches and algorithms to mitigate cold start problems in recommender systems: a systematic literature review. *Journal of Intelligent Information Systems*, *59*(2), 341-366.

[10]. Roy, D., & Dutta, M. (2022). A systematic review and research perspective on recommender systems. *Journal of Big Data*, *9*(1), 59.

[11]. Idrissi, N., & Zellou, A. (2020). A systematic literature review of sparsity issues in recommender systems. *Social Network Analysis and Mining*, *10*(1),15.

[12]. Saxena, D., & Cao, J. (2021). Generative adversarial networks (GANs) challenges, solutions, and future directions. *ACM Computing Surveys (CSUR)*, *54*(3), 1-42.

[13]. Girin, L., Leglaive, S., Bie, X., Diard, J., Hueber, T., & Alameda-Pineda, X. (2020). Dynamical variational autoencoders: A comprehensive review. *arXiv preprint arXiv:2008.12595*.

[14]. Jaweed, M., & Shaikh, R. F. (2024). OPTIMIZING GENERATIVE AI BY OVERCOMING STABILITY MODE COLLAPSE AND QUALITY CHALLENGES IN GANS AND VAES. *MSW Management Journal*, *34*(2), 497-507.

[15]. Rahali, A., & Akhloufi, M. A. (2023). End-to-end transformer-based models in textual-based NLP. *AI*, *4*(1), 54-110.

[16]. Bond-Taylor, S., Leach, A., Long, Y., & Willcocks, C. G. (2021). Deep generative modelling: A comparative review of vaes, gans, normalizing flows, energy-based and autoregressive models. *IEEE transactions on pattern analysis and machine intelligence*, *44*(11), 7327-7347.